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ONCOLOGY

Neoadjuvant proton beam irradiation vs. adjuvant ruthenium brachytherapy in transscleral resection of uveal melanoma

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Abstract

Background Uveal melanoma is the most common primary ocular malignancy in adults in the USA and Europe. The optimal treatment of large uveal melanoma is still under debate. Radiation therapy has its limitation due its eye-threatening secondary complications and is therefore often combined with surgical excision of the tumor.

Methods In a retrospective interventional review, we evaluated in total 242 patients with uveal melanoma that underwent transscleral tumor resection with a predefined protocol, either with adjuvant ruthenium brachytherapy (Ru-106 group, *n* 136,), or with neoadjuvant proton beam therapy (PBT group, *n* 106). Kaplan-Meier estimates with log-rank test were used to show survival curves and a multivariable Cox regression model was used to calculate adjusted rate ratios.

Results Local tumor recurrence rates after 3 and 5 years were 4% (95% CI 1.2–17.8%) and 9.1% (95% CI 2.9–27.3%), respectively, in the PBT group and 24.6% (95% CI 15.8–37.1%) and 27.5 (95% CI 17.8–41.1%), respectively, in the Ru-106 group. This leads to an overall recurrence rate almost 4 times higher in the Ru-106 group compared to the PBT group. After adjusting for the a priori confounders and the tumor distance to optic disc and ciliary body infiltration, the adjusted risk of tumor recurrence was 8 times (RR 7.69 (2.22–26.06), $p < 0.001$) higher in the Ru-106 group as compared to the PBT group. Three- and 5-year metastatic rates were 23.2% (95% CI 5.6–37.1%) and 31.8% (95% CI 20.7–46.8%), respectively, in the PBT group and 13.2% (95% CI 6.8–24.9%) and 30.3% (95% CI 18.3–47.5%), respectively, in the Ru-106 group. There was no statistically significant difference in the overall metastasis rate between the two groups even after adjusting for possible confounders.

Conclusion Transscleral resection of large uveal melanomas combined with neoadjuvant proton beam therapy leads to a lower local tumor recurrence rate compared to transscleral tumor resection with adjuvant ruthenium brachytherapy. There was no statistically significant difference in the occurrence of rubeosis iridis, neovascular glaucoma, and in the need for enucleation later on.

Keywords Uveal melanoma · Neoadjuvant proton beam radiotherapy · Adjuvant ruthenium brachytherapy · Transscleral resection

Introduction

Uveal melanoma is the most common primary ocular malignancy in adults in the USA and Europe. The optimal treatment of large uveal melanoma is still a topic of controversy. The prospective randomized Collaborative Ocular Melanoma Study (COMS) showed that there is no difference in survival between eye-conserving treatment with I-125 brachytherapy and enucleation (COMS Study Group) [1].

Eye-conserving treatment largely consists of radiation therapy with Ru-106, I-125, or proton beam radiotherapy (PBT). These treatments, however, have practical limitations: radiation with Ru-106 brachytherapy is limited to a tumor with a thickness less than 7 mm, I-125 increases the risk of secondary changes throughout the eye [2, 3], and PBT is associated with eye-threatening secondary changes when used on large uveal

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melanomas if not combined with surgical excision of the tumor tissue [4–6]. So far, the decision on how to treat uveal melanomas has largely been based on these practical limitations.

Surgical tumor resection techniques were developed by Stallard and Müller in the 1950s [7, 8]. Later, these surgical treatment approaches were used to overcome some of the limitations of radiotherapy that had emerged in the meantime. The approach in combining excision surgery and adjuvant radiotherapy primarily aimed in reducing the incidence of tumor recurrence. Adjuvant ruthenium brachytherapy has shown to reduce the risk of local recurrence after transscleral resection [9, 10]. Kivela et al. reported in a case-control study a local recurrence rate of about 35% for solitary TSR in contrast to 15% if combined with a 20-mm ruthenium plaque [11]. Neoadjuvant proton beam irradiation has been found to reduce further the probability of local tumor recurrence [12].

In this study, we analyzed data on 242 patients with large uveal melanomas that had been treated with transscleral resection. Patients were treated with TSR and either adjuvant ruthenium brachytherapy or neoadjuvant proton beam therapy and followed up for up to 10 years. We compared local tumor recurrence rates and metastasis rates between both treatment arms to explore if the different treatments affected local and metastatic tumor control.

Patients and methods

This study was in accordance with the tenets of the Declaration of Helsinki and was approved by the institutional review board of the Charité Universitätsmedizin Berlin, Berlin, Germany. In a retrospective interventional chart review, we evaluated in total 242 patients from whom 136 patients that were treated by TSR and adjuvant ruthenium brachytherapy (Ru-106 group) and 106 patients who were treated with neoadjuvant proton beam therapy (PBT group) and subsequent TSR. All patients were submitted to TSR under systemic hypotension as described below.

Patients with adjuvant Ru-106 were treated between September 1993 and August 2004. After 2004, Ru-106 brachytherapy was replaced by neoadjuvant proton beam therapy prior to TSR.

Preoperative assessment and eligibility criteria

All patients underwent a complete ophthalmological examination. On this occasion, an ultrasound examination was performed and the largest basal tumor diameter (LBD), tumor thickness, and the distances from the posterior tumor margin to the optic disc (TDO) and fovea (TDF) were measured. Infiltration of the ciliary body was verified by transillumination. Inclusion criteria for TSR were a minimum tumor thickness of 7.0 mm irrespective of ciliary body or anterior chamber invasion, the posterior tumor margin reaching no closer than 3 mm to the optic disc and/or fovea, and a LBD of < 23 mm. Anterior chamber involvement was no exclusion criteria.

Only patients with a preoperative visual acuity greater than 0.05, irrespective of the status of the fellow eye, were offered this procedure.

To identify possible unknown risks for the systemic hypotension during surgery, all patients underwent a medical check-up. This included 24-h blood pressure monitoring, a 24-h electrocardiogram (ECG), a general neurological examination, an electroencephalogram (EEG), and a Doppler ultrasound examination of the external carotid arteries to exclude any clinically relevant stenosis. Additionally, abdominal ultrasound was performed to rule out liver metastasis.

Surgical techniques

Transscleral excision was performed under systemic arterial hypotension as previously described by Foulds [13, 14] and others [15] with modifications [9, 16]. In short, after demarcation of the peripheral tumor borders by transillumination, the sclera was dissected in a lamellar manner, maintaining a scleral flap of more than 90% thickness. Thereafter, the IOP was lowered by a partial one port vitrectomy and the suprachoroidal space was opened with blunt scissors avoiding injury of the choroid. The tumor was gently excised by choroidectomy, avoiding injury to the retina and the scleral flap sutured back. After the resection, in most cases, a SF6 gas bubble was injected in the vitreous cavity to avoid postsurgical hypotony and excessive bleeding.

Irradiation

In the Ru-106 group, a 20.0-mm Ru-106 plaque (CCB; BEBIG, Berlin, Germany) was placed on the excision bed directly after tumor excision. The prescription point for plaque dosimetry was defined in all cases as the apex of an imaginary 5.0-mm thick tumor receiving 100 Gray. The mean scleral dose for this adjuvant brachytherapy was 470 Gray, ranging between 400 and 500 Gray.

In contrary, the PBT group was irradiated prior to TSR in a neoadjuvant fashion. For this purpose, four tantalum rings were placed at the sclera under general or local anesthesia. In 94 patients, treatment planning was performed based on orbital MRI with the OCTOPUS software [17, 18] and then checked by tumor delineation in fundus photography and ultrasound measurements of the tumor and the tantalum rings. In the rest of the group, the eye model was developed on computer tomography scans and the target volume was calculated

Table 1 Baseline characteristics of 140 matched patients separated by treatment arm

	TSR + PBT	TSR + Ru106	*p*-value
Sex			
Female	37 (52.9)	42 (60.0)	0.394
Male	33 (47.1)	28 (40.0)	
Age in years			
<50	20 (28.6)	26 (37.1)	0.281
≥50	50 (71.4)	44 (62.9)	
Mean (SD)	57 (12)	50 (12)	
LBD			
< 16mm	34 (48.5)	37 (52.9)	0.612
≥ 16mm	36 (51.6)	33 (47.1)	
Mean (SD)	15.9 (2.6)	15.7 (2.6)	
TDO*			
< 10mm	67 (95.7)	57 (87.7)	0.089
≥ 10mm	3 (4.3)	8 (12.3)	
Mean (SD)	9.3 (4.5)	9.6 (3.9)	
TDF **			
< 10mm	38 (54.3)	26 (48.1)	0.498
≥ 10mm	32(45.7)	28 (51.9)	
Mean (SD)	9.1 (4.8)	9.7 (4.5)	
Histological type			
Spindle cell	33 (47.1)	33 (47.1)	
Epitheloid	37 (52.9)	37 (52.9)	
Mixed	0 (0)	0 (0)	
Retinal detachment			
No	56 (80.0)	49 (70.0)	0.172
Yes	14 (20.0)	21 (30.0)	
Extra scleral tumor extension			
No	70 (100)	70 (100)	
Yes	0 (0)	0 (0)	
Tumor thickness (mm)			
≤ 9.5	23 (32.9)	21 (30.0)	0.716
> 9.5	47 (67.1)	49 (70.0)	
Mean (SD)	10.4 (1.7)	10.3 (1.8)	
Ciliary body infiltration			
No	13 (19.0)	13 (19.0)	
Yes	57 (81.0)	57 (81.0)	

Missing observations in *5, **6

based on intraoperative measurements of the tantalum clips to the tumor. With these results, the treatment plan was calculated with the EYEPLAN software [19].

The dose was planned to reach 50% of the prescribed dose of 54.5 Gray in a distance of 2.5 mm around the defined outlines of the clinical target volume. This margin was chosen due to the degree of uncertainty in positioning the patient, the penetration depth of the proton beam, and laterally to compensate for the penumbra of the collimator adjusting the proton beam. The relative radiobiological effectiveness of 1 Gray translates into 1.1 Cobalt Gray Equivalent (CGE) [20]. The total dose of 54.5 Gray was divided into four sessions of 15 CGE each.

Postoperative assessment

The first follow-up examination was set to be 1 month after TSR, with a range from a few days to 6 months. In the first 2 years of follow up, examinations were performed every 3–12 months. In the period until 5 years after the initial treatment, patients were seen every 6–12 months. Thereafter, the intervals between examinations ranged from 6 to 24 months depending on the need to treat secondary changes.

Significant areas of pigmented tumor that did not resolve spontaneously in the first 3 months were considered to be residual tumor. Tumor recurrence was defined as any increasing pigmented intraocular lesion or progressive residual tumor

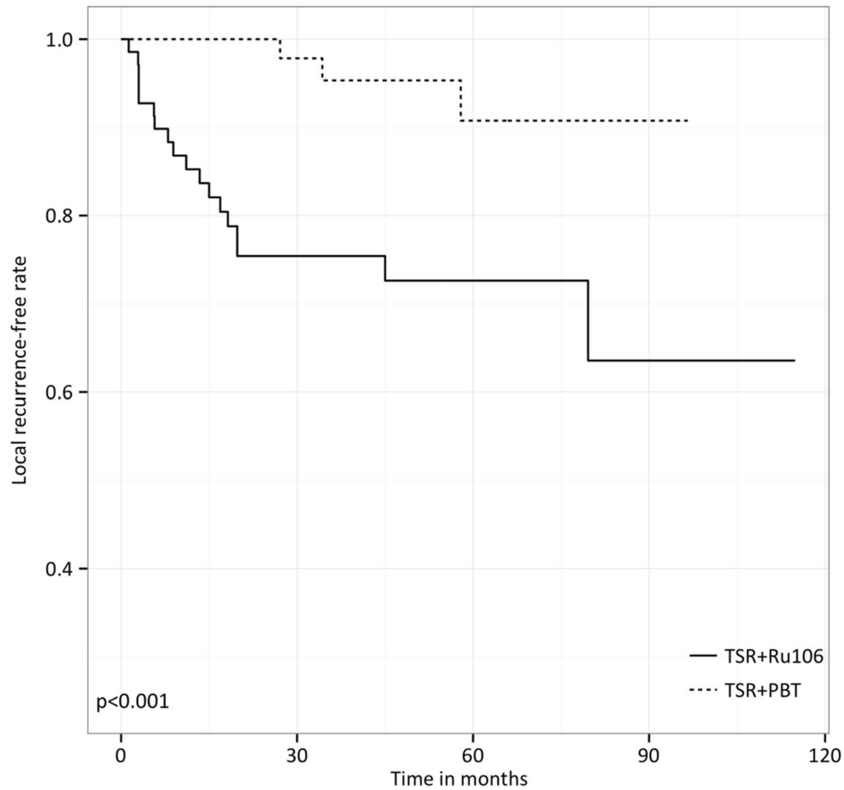

Fig. 1 Local recurrence-free rate with log-rank test; overall and stratified by larger basal tumor diameter

that was not treated in the first 3 months after operation. Residual tumors that did not resolve over time and recurrent tumors received additional treatment. This could either be treatment intended to retain the eye including photocoagulation, transpupillary thermotherapy (TTT), additional Ru-106 plaque brachytherapy (in the Ru-106 group), and external beam radiotherapy or enucleation and exentaration.

All patients were advised to undergo liver ultrasound every 3–6 months for detection of metastasis.

Postoperative complications such as vitreous or subretinal hemorrhages, extended retinal detachments, and radiotherapy-induced proliferative retinopathies were treated conventionally by vitreoretinal surgery (buckling procedures and/or pars plana vitrectomy) and photocoagulation. Secondary cataract was removed with phacoemulsification and an intraocular lens was implanted. For eyes which developed phthisis, enucleation was offered to the patient.

Statistical methods

Data was entered into a database and checked for data entry errors using frequency checks. Continuous variables were categorized according to predefined values (REF). Missing data was excluded from the analysis. To achieve comparable groups, we matched with a 10% error margin on LBD size and tumor height as well as absolutely tumor cell type.

The study windows were synchronized to a 10-year follow-up period starting the day of diagnosis. To model the relationship between treatment arms and local tumor recurrence as well as treatment arm and metastatic rate, we used a Cox proportional hazard, after exploring the proportional hazard assumption. For the analysis of metastasis, tumor recurrence was modeled as a time-dependent variable.

The model building process started by exploring the relative distribution of possible confounders (except for the matching variables) between treatment arms using two by two tables and a chi square test. We then calculate the crude local tumor recurrence and metastasis rate and the respective rate ratios. The crude rate ratios were adjusted by each possible confounding variable using the Mantel-Haenszel methods. Variables that changed the crude rate ratio by approximately 10% were identified as confounders. The test for homogeneity of stratum-specific rate ratios was observed to identify effect modification.

All confounding variables, including the a priori confounders age and sex, were entered into a multivariable Cox regression model to calculate adjusted rate ratios and 95% confidence intervals. Variables were entered in a stepwise forward manner starting with the strongest confounders. Interaction terms were fitted for variables with a possible effect modification and tested using a Wald test. Stratum-specific rate ratios were presented where appropriate. Kaplan-Meier estimates with log-rank test were used to show survival curves.

Data on visual acuity was plotted using boxplots and analyzed using the Mann-Whitney *U* test or Wilcoxon signed

Table 2 Tumor recurrence in 140 matched patients treated at the Charité from 1993 until 2010

	n/N	10-year rate in % (95% CI)	Adjusted hazard ratio (95% CI)	p value
TSR + PBT	3/70	9.1 (2.8–27.3)	1	0.001
TSR + Ru-106	18/70	36.5 (20.7–59.1)	7.69 (2.22–26.06)	
Sex				
Female	8/79	13 (4.5–25.5)	1	0.171
Male	13/61	32.5 (17.8–54.4)	1.88 (0.75–4.66)	
Age				
< 50	7/46	16.3 (8.1–23.3)	1	0.447
≥ 50	14/94	30.1 (14.8–56.8)	1.44 (0.51–3.33)	
LBD (mm)				
< 16	7/71	15.2 (6.9–31.6)		
≥ 16	14/69	33.4 (16.4–60.3)		
TDF (mm)				
< 10	9/64	16.7 (8.7–28.9)		
≥ 10	9/60	31.2 (14.2–61.9)		
TDO (mm)				
< 10	19/124	20.6 (13.032.37)	1	0.173
≥ 10	1 /11	9.1 (1.3–49.2)	0.24 (0.03–1.86)	
Histological type				
Spindle cell	9 /66	19.3 (9.5–36.8)		
Epitheloid	12/74	34.8 (13.7–71.1)		
Retinal detachment				
No	16/105	18.8 (11.7–29.6)		
Yes	5/35	30.1 (11.3–67.7)		
Extra scleral tumor extension				
No	21/140	24.4 (14.0–40.0)		
Tumor thickness (mm)				
≤ 9.5	7/44	32.1 (13.9–63.1)		
> 9.5	14/96	16.9 (10.1–27.2)		
Ciliary body infiltration				
No	3/26	15.9 (5.0–44.4)	1	0.392
Yes	18/114	25.1 (14.1–42.1)	1.73 (0.49–6.11)	

rank test, where appropriate. All statistical analyses were performed using STATA version 12.1 (StataCorp, College Station, Texas, USA). The survival package in R software (R Core Team, Vienna, Austria) version 2.15.1 was used to generate Kaplan-Meier plots.

Results

In total, 242 patients were recruited out of which 140 matched pairs were created. The remaining patients were dropped from the analysis. Baseline characteristics are shown in Table 1.

Local tumor control and tumor recurrence

The median follow-up time was 34.4 months (range 0.8–120). A total of 21 tumor recurrences were noted, giving an overall recurrence rate of 24.3% (95% CI 14.0–40.1%). The 3- and 5-year tumor recurrence rate was 14.8% (95% CI 9.5–22.6%) and 18.6% (9% CI 12.0–28.9%), respectively.

In the PBT group, 3- and 5-year recurrence rates were 4% (95% CI 1.2–17.8%) and 9.1% (95% CI 2.9–27.3%), respectively, and 24.6% (95% CI 15.8–37.1%) and 27.5% (95% CI 17.8–41.1%), respectively, in the Ru-106 group. The overall recurrence rate was almost 4 times higher in the Ru-106 group compared to the PBT group (Fig. 1, Table 2.)

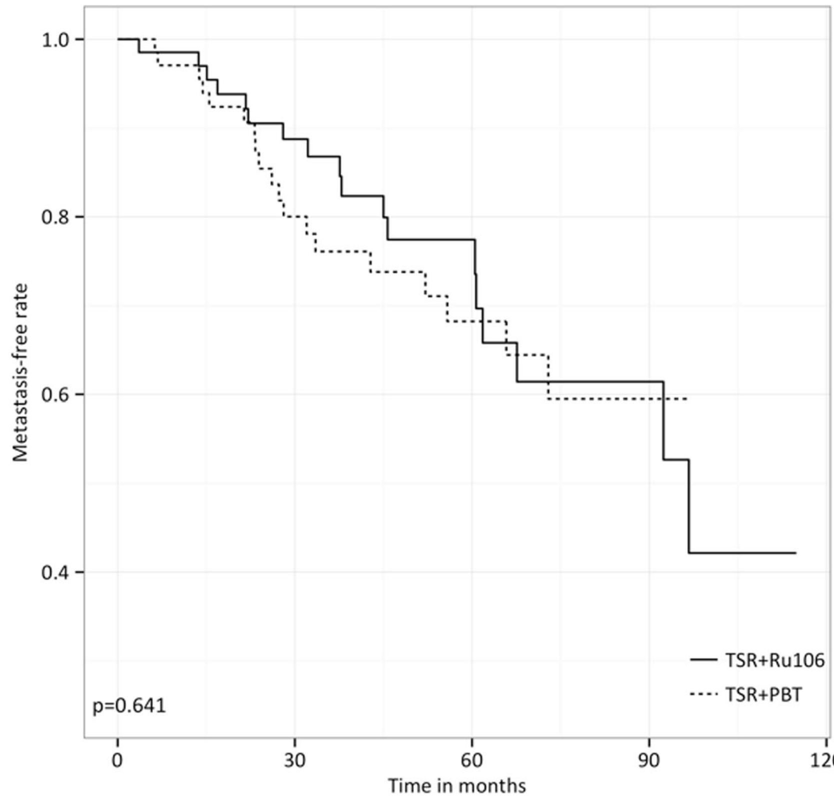

Fig. 2 Metastasis-free rate with log-rank test

Besides the a priori confounder age, sex, and tumor cell type, TDO and ciliary body infiltration confounded the association between treatment arm and tumor recurrence. When adjusted for these confounders, the likelihood of tumor recurrence in the Ru-106 group was almost 8 times higher than in the PBT group (Table 2).

Systemic tumor control: tumor-specific survival and metastasis

The median follow-up time was 39.8 months (range 0.8–120). A total of 37 tumor metastasis were noted, giving an overall metastasis rate of 54.4% (95% CI 35.7–75.4%). The 3- and 5-year tumor recurrence rates were 18.5% (95% CI 12.6–26.9%) and 30.1% (9% CI 22.3–41.7%), respectively.

In the PBT group, 3- and 5-year metastasis rates were 23.2% (95% CI 5.6–37.1%) and 31.8% (95% CI 20.7–46.8%), respectively, and 13.2% (95% CI 6.8–24.9%) and 30.3% (95% CI 18.3–47.5%), respectively, in the Ru-106 group. There were no differences in the overall metastasis rate between PBT and Ru-106 groups (Fig. 2, Table 3.)

For the analysis of the adjusted metastasis rate, we identified the same variables as confounders as in the analysis of the recurrence rate. After adjusting for confounding, no differences in the metastasis rate were found between PBT and Ru-106 (Table 3).

Visual acuity

At baseline, visual acuity was similar in both treatment arms (median 0.4 [IQR 0.2–0.7] in PBT versus 0.3 [IQR 0.1–0.7] in Ru-106, $p = 0.031$) (Fig. 3). One year post intervention, visual acuity dropped to 0.8 (IQR 0.5–1.3) in the PBT group ($p < 0.001$) and to 1.5 (IQR 1–2) in the Ru-106 group ($p < 0.001$). Two years post intervention, visual acuity in the PBT group had further worsened (1.2 [IQR 0.8–1.5], $p < 0.001$) but improved in the Ru-106 group (0.8 [IQR 0.4–1.2], $p < 0.001$). Throughout the rest of the study period, visual acuity was better in the Ru-106 group than in the PBT group although no significant difference was noted beyond 6 years of follow-up (Fig. 3).

Secondary complications

Over the course of the study period, a total 17 eyes were enucleated: 6 (8.5%) in the PBT group and 11 (15.7%) in the Ru-106 group ($p = 0.196$). Rubeosis of the iris was seen in 1 (1.4%) and 0 of the patients in the PBT and Ru-106 group ($p = 0.316$), respectively. Neovascular glaucoma occurred in 1 (1.4%) of the patients in the PBT group and Ru-106 group.

Table 3 Tumor metastasis in 140 matched patients treated at the Charité from 1993 until 2010

	n/*N*	10-year rate in % (95% CI)	Adjusted hazard ratio (95% CI)	*p* value
TSR + PBT	19/70	40.1 (26.6–58.6)	1	0.884
TSR + Ru-106	18/70	56.9 (34.9–80.8)	0.951 (0.48–1.86)	
Local tumor recurrence				
No	29/119	53.3 (29.3–81.3)	1	0.273
Yes	8/21	64.4 (33.7–92.5)	1.65 (0.67–4.04)	
Sex				
Female	21/79	41.5 (27.6–48.8)	1	0.353
Male	16/61	59.7 (33.9–86.4)	0.71 (0.35–1.44)	
Age				
< 50	10/46	34.5 (19.4–56.4)	1	0.124
≥ 50	27/94	67.1 (40.8–90.5)	1.81 (0.85–3.83)	
LBD (mm)				
< 16	20/71	59.8 (30.4–89.9)		
≥ 16	37 /69	48.3 (28.5–72.6)		
TDF (mm)				
< 10	12/64	34.6 (19.2–57.2)		
≥ 10	20/60	55.6 (36.2–77.1)		
TDO (mm)				
< 10	33/124	20.6 (13.032.37)	1	0.234
≥ 10	1/11	9.1 (1.3–49.2)	0.41 (0.09–1.77)	
Histological type				
Spindle cell	17 /66	56.9 (30.1–86.2)		
Epitheloid	20/74	54.2 (29.7–82.2)		
Retinal detachment				
No	27/105	43.3 (30.6–58.8)		
Yes	10/35	71.5 (32.2–98.3)		
Extra scleral tumor extension				
No	37/140	54.5 (35.7–75.5)		
Tumor thickness (mm)				
≤ 9.5	9/44	42.7 (20.1–74.8)		
> 9.5	28/96	58.3 (34.9–83.2)		
Ciliary body infiltration				
No	6/26	40.1 (18.0–75.3)	1	0.689
Yes	31/114	55.8 (35.7–77.9)	1.21 (0.48–2.96)	

Discussion

Our results show that additional treatment with either neoadjuvant proton beam therapy or adjuvant Ru-106 brachytherapy affects the incidence of local recurrence in patients with large uveal melanoma undergoing transscleral resection.

The results of this study should be interpreted with caution. Our baseline analysis shows considerable difference between the two treatment arms, possibly affecting recurrence and metastasis rates. We addressed the issue by identifying matched pairs for the most important confounders. This matching resulted in comparable groups although some differences remained. It is therefore possible some degree of residual confounding affected our analysis. Also, it is important to notice that as a result of matching, we had to exclude a considerable number of patients and hence our results are only applicable to the remaining matched pairs.

Another source of bias could have arisen from the different study periods during which patients were recruited and treated. Beyond the new treatment with PBT instead of Ru-106, the treatment and care of large uveal melanomas remained the same over the years.

Considering local tumor recurrence, our data suggest that PBT treatment achieves greater tumor control than Ru-106. This effect increases with the basal tumor diameter. Reasons for the higher risk of local tumor recurrence in the adjuvant Ru-

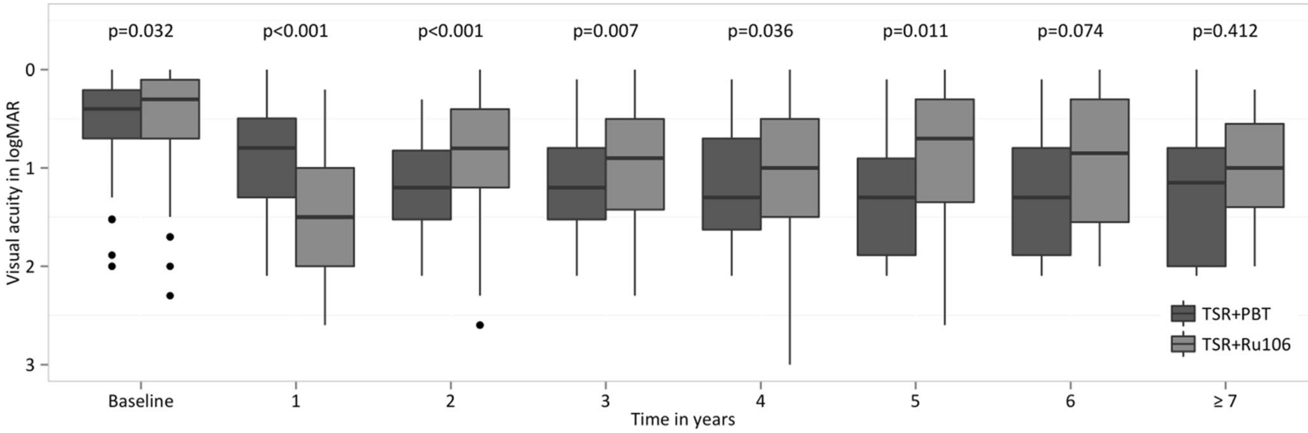

Fig. 3 Visual acuity comparing patients treated with PBT and Ru-106 during follow-up period

106 group could be that in some cases the tumor margins might not be entirely covered with an adequate safety margin by the ruthenium plaque. Damato already expressed a preference for a 25-mm ruthenium plaque to add a wider field of radiation [21] in tumors with a large basal diameter. We preferred to address this issue by switching to neoadjuvant proton beam irradiation before performing transscleral resection after 2004.

It is crucial to consider that failing to locally control the tumor may increase the risk of metastasis. A recent study in patients with uveal melanoma could show that local recurrence increases the risk of metastasis [9, 22]. This present study is in accordance with these findings.

Besides local and metastatic tumor control, secondary complications are important to consider in the treatment of uveal melanoma. PBT therapy is known for its complications such as "toxic tumor syndrome" [21, 23]. Our findings show no higher rate of secondary complications as rubeosis iridis and neovascular glaucoma. Toxic tumor syndrome is described to take place between 1 and 2.5 years after treatment [24]. In our study, the tumors on the PBT group were excised shortly after radiation and therefore prior to the toxic tumor mass being able to develop those secondary changes. There were also no differences with regard to eye retention between the adjuvant ruthenium and neoadjuvant proton beam treatment groups.

Nevertheless, the rate of secondary complications in patients with large uveal melanoma treated only by irradiation without resection of the tumor is significantly higher [11]. TSR is therefore an adequate means to overcome practical limitations of different radiotherapy options. By introducing neoadjuvant radiotherapy, it was possible to reduce significantly the incidence of local tumor recurrence without a higher rate of secondary complications.

In the course of the study, differences in visual acuity outcomes tended to diminish between patients treated with adjuvant ruthenium or neoadjuvant proton beam irradiation options. It is therefore reassuring that the advantage of reducing significantly the incidence of local tumor recurrence by neoadjuvant proton beam irradiation has no dismal long-term consequences on visual function in these patients.

In conclusion, the majority of patients treated with PBT will achieve good tumor control and retain some useful visual function.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee (name of institute/committee) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Human and animal rights and informed consent For this type of study, formal consent is not required.

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